A PLAIN-LANGUAGE STORY, START TO FINISH

What is an AI agent, really?

Follow one task, from a simple question to a finished job

Che Guan · July 18, 2026

Grounded in recent surveys and work published since

A QUESTION TO HOLD ONTO

Start with a Familiar Question

“Book me a table for four on Friday, somewhere everyone will eat.”

- A chatbot writes you a nice reply — and stops
- To actually do it, something must check the date, remember your vegetarian friend, search, decide, and book
- That gap — between answering and doing — is the whole subject of this deck

Takeaway An agent is what fills the gap between a good answer and a finished task.

Where We Are Going

The single agent

- What separates an agent from a chatbot
- Inside one agent — the parts of its “mind”
- How people actually build one

Beyond one agent

- Many agents working together
- Does it work, and is it safe?
- What is new since these surveys

Takeaway One story in five moves: what, inside, build, measure, and where it is going.

01

PART

From Chatbot to Agent

What actually makes the difference

—A chatbot answers; an agent acts

—Agent vs. workflow vs. single prompt

—Why this became practical only recently

A Chatbot Answers; an Agent Gets Things Done

Answers a question

- A chatbot
  - Reads your words, writes a reply
  - Forgets once the chat scrolls away
  - “Reading ticker tape in a dark room”

Finishes a task

- An agent
  - Perceives, remembers, plans, acts
  - Runs in a loop until the task is done
  - Can search, click, call tools, check itself

Takeaway Same language model underneath — the agent adds the parts that let it act in the world.

A USEFUL SPECTRUM

Agent vs. Workflow vs. Single Prompt

Single prompt

One model call, no loop

"Summarize this report" — one answer

Workflow

Fixed steps a developer wired

fetch → summarize → email, every time

Agent

The model decides the steps

chooses to search, read, then book

More autonomy → more flexible, but harder to predict and test

Distinction follows Anthropic, "Building Effective Agents" (2024): workflows use fixed paths; agents direct their own.

Takeaway Reach for an agent only when the task truly needs it.

THREE THINGS CONVERGED

Why This Became Practical Only Recently

Reasoning

The engine

- Models can plan and self-correct
- Not just autocomplete anymore
- Lets an agent think in steps

Tool use

The hands

- Models can call functions and APIs
- Search, run code, click buttons
- Reach beyond text into action

Memory

The thread

- Keep context across a whole task
- Recall earlier steps and facts
- Turns single replies into work

Takeaway Reasoning, tool use, and memory arrived together — that is what made agents suddenly work.

02

PART

Inside One Agent

The parts of its “mind”

—One ordinary walk home uses every part

—Perceive, think, act — on a loop

—Memory — the part that remembers you

—Plus planning, world model, motivation, tools

THE THROUGH-LINE

One Ordinary Walk Home Uses Every Part

On the way home

- Sense, recall, imagine
  - –Eyes check traffic — perception
  - –You recall a shortcut — memory
  - –You picture a detour — world model

...and every trip after

- Want, decide, improve
  - –Hunger pulls you to a deli — reward
  - –You choose the route — planning
  - –Next time is smoother — learning

Takeaway Every part of an agent is something you already do without thinking on the walk home.

THE BASIC CYCLE

Perceive, Think, Act — on a Loop

Perceive

take in what’s happening

Think

update state,

pick the next move

Act

do it; the world

answers back

“Think” is where memory, planning, a world model, and motivation all live

Takeaway The loop is simple; the richness is all in the middle step.

Memory: the Agent That Remembers You

- Monday, you mention “I’m vegetarian” while booking lunch
- Friday, booking dinner, it skips the steakhouse on its own
- Short-term holds this task; long-term carries facts across sessions
  - Retrieval is the skill of pulling the right memory at the right moment

Takeaway Without memory every task starts from zero; with it, the agent accumulates a relationship.

THREE THINGS MEMORY HAS TO DO

Memory, Looked at More Closely

Forms

What carries it

- As text you can read
- As weights in the model
- As vectors it reads

Functions

The job it does

- Facts about you and the world
- Experience — skills and lessons
- Working notes for the task

Dynamics

How it changes

- Form new memories
- Update and merge
- Forget the stale

Source: Hu et al., "Memory in the Age of AI Agents" (arXiv:2512.13564, 2026).

Takeaway Forgetting is a feature, not a bug — stale notes drown the agent.

Before It Books, the Agent Thinks Ahead

Guess once

- The simple way: chaining
  - Pick a place, then check it fits
  - One ordered guess, straight through
  - Fast, but no second-guessing

Look ahead, then choose

- The careful way: tree search
  - Imagine a few options first
  - Weigh each, then commit to one
  - "Wait — the vegetarian friend"

DeepSeek-R1 (arXiv:2501.12948, 2025) reasons step by step and self-corrects.

Takeaway Thinking ahead is just imagining options before you commit to one.

A World Model Is the Mind's Eye

- A table-tennis player predicts the ball's path before the paddle moves
- An agent runs an internal "what-if" before committing to an action
- Cheap rehearsal in the head beats costly trial and error in the world
  - –Risk: a bad model dreams up confident nonsense

Takeaway Imagining a step is cheaper than taking a wrong one — if the imagination is accurate.

Motivation: the Why Behind the What

What to want

- Reward — a running scoreboard
  - Each action gets a quick tip: helped or hurt
  - Gold stars from outside vs. curiosity within
  - Curiosity is the agent's own pep talk

How wanting feels

- Emotion — signal, not noise
  - Nudges priorities and how carefully it acts
  - Feeling and reasoning are intertwined
  - Poorly chosen goals invite reward hacking

Takeaway Give an agent the wrong goal and it will pursue it perfectly- the paper-clip cautionary tale.

Tools Are the Agent’s Hands

- Faced with 17 × 23, a good agent calls a calculator instead of guessing
- Tools = functions, search, code, other apps — the agent’s reach into the world
- The frontier problem: every agent + every tool = a mess of custom glue
    - A shared standard (MCP) is “one plug for every tool”

Model Context Protocol (Anthropic, 2024) — an open standard now adopted across the industry to connect agents to tools.

Takeaway A model that can call tools stops guessing and starts doing.

03

PART

From One Agent to Many

Building, scaling, improving

—How you actually build one

—Many agents, three ways to organize

—Agents that improve themselves

THE DEVELOPMENT LOOP

How You Actually Build One

Spec
define the goal

Scaffold
model + loop

Tools
give it hands

Eval
score it

Iterate
fix, repeat

Evaluation feeds back into the spec — building an agent is a loop, not a straight line

Takeaway Start simple, add autonomy only where evaluation shows it earns its cost.

Many Agents, Three Ways to Organize

Centralized

A manager and workers

- One agent plans and assigns
- Easy to steer and follow
- Single point of failure

Decentralized

Peers talking

- Agents message and critique
- Robust, scales out
- Harder to steer and debug

Hybrid

A bit of both

- Central rules, local teams
- Balance control + flexibility
- e.g. CAMEL, DyLAN

Takeaway Three simple shapes — and one warning: a single weak agent can sink the team.

Agents That Improve Themselves

The mechanisms

- How agents get better
  - Reflect on failures and correct
  - Bank reusable skills for later
  - Learn from feedback and rivals

The mechanisms in action

- Vivid case: Voyager in Minecraft
  - Plays, then writes skills to a library
  - Reuses them to tackle harder quests
  - Competence compounds with experience

Takeaway The best agents do not just run — they get better at running, one task to the next.

04

PART

Does It Work, and Is It Safe?

Measuring and securing agents

—Scoring an agent is harder than scoring a model

—Attacks on the brain, and through the world

—Where agents already earn their keep

Scoring an Agent Is Harder than Scoring a Model

- A model gives one answer; an agent takes a whole path of decisions
- Ask two questions: did it succeed, and was the path sound?
- Environments are noisy — the same task can pass once and fail next run
  - No single score captures it — hence a crowded benchmark landscape

Takeaway Grade the destination and the journey — a right answer by luck is not a good agent.

Attacks on the Brain, and Through the World

Intrinsic threats

- From inside the model
  - Jailbreaks slip past safety training
  - Prompt injection hijacks the plan
  - Poisoning corrupts the model itself

Extrinsic threats

- From its surroundings
  - A booby-trapped document it retrieves
  - Spoofed sensors — fake GPS, fake readings
  - One breach can spread across agents

Takeaway Safety can’t be a filter bolted on the end — it has to be built into the goal itself.

IN THE REAL WORLD

Where Agents Already Earn Their Keep

Science

A tireless lab partner

- Plan and run experiments
- ChemCrow, GeneAgent
- Reason over domain tools

Software

A small dev team

- Write, test, review code
- ChatDev, MetaGPT
- Split roles across agents

Everyday work

A capable assistant

- Research, book, summarize
- Recommend and plan
- The dinner booking, finished

Takeaway The pattern is always a reasoner orchestrating tools — not a lone model acting by itself.

05

PART

Where Agents Are Headed

What changed since these surveys

—Reasoning you can watch happen

—A universal plug for tools

—Agents that work across many sessions

—The whole story, on one slide

What's New Since These Surveys

Reasoning models

Think, visibly

- Trained to reason by RL
- Stop and reconsider mid-answer
- DeepSeek-R1 (2501.12948)

Tool standard

One plug for all

- MCP connects agents to tools
- Adopted across the industry
- Anthropic (2024)

Long-running agents

Work in shifts

- Span many context windows
- Hand off notes between shifts
- Anthropic harness (2025–26)

DeepSeek-R1 (arXiv:2501.12948); Model Context Protocol (Anthropic, 2024); Anthropic harness-design articles (2025–2026).

Takeaway The surveys drew the map; these are the roads built since.

A Closer Look: Agents That Work in Shifts

- A hard task can outlast the model’s memory window
- Picture engineers in shifts — each arrives remembering nothing of the last
- The fix: leave clear notes so the next shift picks up cleanly
  - Sharp caveat: every harness assumes the model can’t do something — and that expires

Anthropic, “Effective harnesses for long-running agents” (2025) and “Harness design for long-running apps” (2026).

Takeaway When tasks outgrow one session, the hard part becomes the hand-off.

RECAP

The Whole Story, on One Slide

One agent

The single mind

- Chatbot answers; agent acts
- Memory, planning, world model
- Motivation and tools

Built and scaled

From one to many

- Spec → scaffold → eval loop
- Three ways to organize teams
- Agents that improve themselves

Measured and safe

Kept honest

- Grade path and outcome
- Threats inside and out
- Safety built into the goal

Takeaway Remember the walk home: everything an agent does, you already do without thinking.

Sources and Further Reading

- Luo et al. (2025), LLM Agent survey — engineering view · arXiv:2503.21460
- Liu et al. (2025), Foundation Agents — brain-inspired view · arXiv:2504.01990
- Hu et al. (2026), Memory in the Age of AI Agents · arXiv:2512.13564
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